

Stella (Yaxuan) Mao | Curriculum Vitae — September 2026

Human Centered Design & Engineering, University of Washington
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RESEARCH INTERESTS

Human-robot interaction; accessible and assistive robotics; social and educational robotics; robot behavior and embodiment; human-centered robot autonomy and teleoperation; human-AI collaboration; participatory and mixed-methods HCI.

EDUCATION

- 2025 – 2027 **M.S. in Human Centered Design & Engineering**, University of Washington, Seattle
GPA: 3.93 / 4.0
Relevant Coursework: User-Centered Design, HCI Theory, Experimental Research Methods, Usability Testing
- 2020 – 2024 **B.Sc. in Creative Media**, City University of Hong Kong
Cum Laude (Top 10%)
CS Coursework: Computer Programming, Data Structures, Algorithms, AI Game Programming, Multi-modal Interfaces, Computer Vision, Computer Graphics, Machine Learning
Design Coursework: Contemporary & New Media Art, Interactive Narrative, Installation, Game Prototyping, Hardware Hacking, UI/UX Visual Communication

PUBLICATIONS

PEER-REVIEWED JOURNAL & CONFERENCE PAPERS

- 2026 **Yaxuan Mao**, Yanheng Li, Yuzhuo Jia, Fanhong Li, Zi Yin, Shurui Zheng. 2026. *LUMOS: Designing an Interactive Companion Robot to Enhance Critical Reading and Thinking in Academic Contexts*. In Proceedings of the 35th IEEE International Conference on Robot and Human Interactive Communication (RO-MAN).
- 2025 **Yaxuan Mao**, Yanheng Li, Duo Gong, Pengcheng An, Yuhan Luo. “Can I Decorate My Teeth With Diamonds?”: Exploring Multi-Stakeholder Perspectives on Using VR to Reduce Children's Dental Anxiety. In Proceedings of the 28th ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW).
- 2024 Yanheng Li, Long Bai, **Yaxuan Mao**, Xuening Peng, Zehao Zhang, Antoni B. Chan, Jixing Li, Xin Tong, Ray LC. *Affecting Audience Valence and Arousal in 360° Immersive Environments: How Powerful Is Neural Style Transfer?*. In Proceedings of the International Conference on Human-Computer Interaction (HCI).
- 2023 Yanheng Li, Long Bai, **Yaxuan Mao**, Hongliang Ren, Yu Qiao, Xin Tong, Ray LC. *Rethinking Pain Communication of Patients with Alzheimer's Disease through E-textile Interaction Design*. *Frontiers in Human Neuroscience*.
- 2023 Yanheng Li, Long Bai, **Yaxuan Mao**, Xuening Peng, Zehao Zhang, Xin Tong, Ray LC. *The Exploration and Evaluation of Generating Affective 360° Panoramic VR Environments Through Neural Style Transfer*. In Proceedings of the 30th IEEE Conference on Virtual Reality and 3D User Interfaces (VR).

WORKSHOP, LATE-BREAKING, AND WORK-IN-PROGRESS PAPERS

- 2025 Katie Xue, **Yaxuan Mao**, Junnan Yu, Yuhan Luo. *Toward Interactive Reading: Co-designing with Adolescents to Explore Opportunities for Overcoming Reading Challenges*. In Proceedings of the ACM Conference on Interaction Design and Children (IDC), pp. 904-909. 2025.
- 2023 Yanheng Li*, **Yaxuan Mao***, Ray LC. *Communicating Failure Recovery with Robotic Body Movement*. In Proceedings of the 21st IEEE International Conference on Robotics and Automation (ICRA), Workshop. (*equal contribution)

2023 Yanheng Li, Luoying Lin, Xinyan Li, **Yaxuan Mao**, Ray LC. “Nice to Meet You!”: Expressing Emotions with Movement Gestures and Textual Content in Automatic Handwriting Robots. In Proceedings of the 18th ACM/IEEE International Conference on Human-Robot Interaction (HRI), Late-Breaking Report.

MANUSCRIPTS UNDER REVIEW

2026 **Yaxuan Mao**, Yanheng Li, Sourojit Ghosh. *Who Goes First Matters: How Workflow Ordering Structures Automation Bias in Human-AI Collaboration*. Manuscript submitted to the ACM Conference on Human Factors in Computing Systems (CHI), 2026.

2026 Evolone Layne*, **Yaxuan Mao***, Ege Caglar, Maya Cakmak. *Leveling the Playing Field: Enabling Adults with Motor Limitations to Participate in Play via Mobile Manipulator Robots*. Manuscript submitted to the ACM/IEEE International Conference on Human-Robot Interaction (HRI), 2026. (*equal contribution)

2026 Evolone Layne, **Yaxuan Mao**, Maya Cakmak. *Revisiting Privacy Attitudes Toward Teleoperated Robots in the Era of In-Home Humanoids*. Manuscript submitted to the ACM/IEEE International Conference on Human-Robot Interaction (HRI), 2026.

RESEARCH EXPERIENCE

May. 2026 – Present **Human-Centered Robotics Lab (HCR Lab)**, University of Washington
Supervised by Dr. Maya Cakmak

Accessible Teleoperation for Physical Participation in Play

- Conducted a community-based study with adults with severe motor limitations using a teleoperated mobile manipulator for basketball, bowling, and functional tasks.
- Adapted robot-control interfaces for single-switch scanning, head-controlled dwell input, direct touch, and AAC-based access.
- Collected and analyzed observations, interviews, rating scales, and practitioner feedback to examine accessibility, agency, enjoyment, and meaningful participation.

Privacy and Utility in Remotely Teleoperated Home Robots

- Designed and analyzed survey studies examining users’ privacy concerns and preferred privacy protections for remotely teleoperated home robots.
- Investigated how visual filters and household task contexts influence the tradeoff between end-user privacy and teleoperator task performance.
- Conducted literature review, quantitative and qualitative analysis, data visualization, and manuscript development for an HRI publication.

Feb. 2026 – Present **Sidekick & ActivaTeen**, University of Washington
Supervised by Dr. Jessica Jenness and Dr. Elin A. Björling

- Designed semi-structured interview protocols to study adolescents' use of a digital mental-health intervention system.
- Conducted qualitative thematic analysis of interview data to surface design implications.

Oct. 2025 – Present **TELL-Equity**, University of Washington
Supervised by Dr. Elin A. Björling and Dr. Belinda Y. Louie

- Replicated and refactored a legacy Firebase-based social robot platform using Python and a web stack (HTML/JS/CSS).
- Designed and implemented new robot behaviors and features tailored to K–12 educational co-design workshops.

Sep. 2025 – Aug. 2026 **Human-Centered Data Sciences Lab**, University of Washington
Supervised by Prof. Sourojit Ghosh

- Investigated automation bias in human-AI collaboration, examining how protocol ordering (AI-first vs. human-first) shapes reliance on AI recommendations and error types (commission vs. omission).
- Designed an experiment manipulating AI reliability and task conditions to measure decision outcomes and verification behavior.
- Co-authoring a paper targeting the ACM Conference on Human Factors in Computing Systems (CHI).

Apr. 2025 – **Tsinghua Laboratory of Brain and Intelligence**, Tsinghua University

Jan. 2026	- Applied a mixed-methods approach using user interviews and thematic coding alongside usability and performance metrics to inform social robot interface design. - Designed and built a robot prototype by extending an open-source hardware design with custom Arduino functionality, increasing user reading focus (+26.7%) and engagement (+15.9%) in testing.
May. 2024 – Mar. 2025	Synteraction (formerly NUS-HCI Lab), City University of Hong Kong <i>What Is Real? A Survey of AI-Enabled Artifact Creation and Its Misleading Nature</i> - Conducted a comprehensive review of AI-enabled artifact creation, focused on ethical implications and human perceptions of AI-facilitated “ArtiFacts.” - Developed a conceptual framework situating ArtiFacts within AI ethics research.
June. 2023 – Mar. 2025	BiWell, City University of Hong Kong <i>VR-Based Intervention for Dental Anxiety in Children</i> - Led co-design workshops with children, guardians, and dentists to incorporate multi-stakeholder perspectives into a VR system built in Unreal Engine. - Analyzed qualitative workshop data to inform the design and the resulting publication.
Apr. 2023 – Aug. 2023	NUS-HCI Lab, National University of Singapore <i>RDog: A Robotic Guide Dog with Multi-Modal Feedback System</i> - Designed a ROS-based robotic guide dog focused on multi-modal feedback mechanisms. - Conducted user experiments and interviews to refine system usability.
Jul. 2022 – May. 2023	Studio for Narrative Spaces, City University of Hong Kong - Communicating Failure Recovery with Robotic Body Movement: designed non-humanoid robot movement algorithms to non-verbally convey error and recovery strategies, validated through participatory design sessions. - Emotional Expression with Movement and Verbal Patterns of an Automatic Handwriting Robot: studied how handwriting features (baseline, slant, pressure, size, spacing, speed) shape users' emotional responses to a robot. - Affecting Audience Valence and Arousal in Immersive Environments: applied neural style transfer to 360° VR environments and evaluated emotional impact via surveys and interviews. - Dynamize Your Roles, Regulate Your Learning: studied self-regulated learning among content creators on the RED social media platform through surveys, interviews, and quantitative analysis.

PROFESSIONAL EXPERIENCE

Apr. 2025 – Sep. 2025	AI Researcher Intern, Digital China Conducted human-in-the-loop evaluation of LLM agents in enterprise HR scenarios and refined prompts based on user-intent analysis, improving scenario-specific response accuracy from 32% to 86%.
Jun. 2024 – Oct. 2024	UI Designer, Youth Incubator Implemented and maintained UI designs (HTML/CSS/JS) for the organization's website.
Aug. 2021 – Feb. 2023	Co-Founder & Product Manager, SeeWall Limited Founded a startup developing assistive technologies for the visually impaired (barrier-avoidance belt and blind-friendly software), securing HK\$100,000 in seed funding through HK Tech 300.
Jun. 2022 – Aug. 2022	Game Design Intern, Tencent Games Designed and developed game scenarios in Unreal Engine 4.

TECHNICAL SKILLS

Programming: C++, C#, JavaScript, HTML/CSS, Python, Arduino, Processing, SQL, OpenCV, openFrameworks, D3.js, ROS
Design & Prototyping: Adobe Photoshop/Illustrator/Premiere/After Effects, Unreal Engine 5, Unity, Figma, Maya

AWARDS & HONORS

2026	UW Graduate Student Conference Presentation Awards
2026	UW HCDE DEI Committee Mini Grant, University of Washington
2020-2024	HK Tech Tiger (Top 10% of the school)

2023	Special Award, Qianhai Guangdong–Hong Kong–Macao Youth Innovation and Entrepreneurship Competition
2023	Hong Kong Government Scholarship Award
2022	Third Prize, “Chuang Hui Xiang Xiang” Innovation and Entrepreneurship Competition
2021	Seed Fund Award (HK\$100,000), HK Tech 300